EDMS 08-201-3

SPECIFICATION
FOR
DRY-TYPE DISTRIBUTION TRANSFORMERS
FROM 50 kVA UP TO 5000 kVA
22 / 0.4 kV
EPOXY SOLID-CAST / CAST RESIN COIL

Issue: Aug.-2023/ Rev- 3

- This revision contains option items that must be selected by EDC before bidding.

CONTENTS

1. SCOPE

This specification specifies the minimum technical requirements for design, manufacturing, testing, inspection, supply and delivery of three-phase epoxy solid-cast resin coil, two-winding dry-type distribution transformers. The required transformers are step-down transformers to convert the medium voltage level (22 kV) into low voltage level 0.4 kV. The transformers shall be suitable for indoor installations of the power networks of Egyptian Electricity Distribution Companies (EDCs).

2. APPLICABLE STANDARDS

The equipment/material covered in this specification shall comply with the latest edition/amendment of the standards given in Table (1). Where any provision of this specification differs from those of the standards listed below, the provisions of this specification shall apply. Any deviations from the listed standards or the provisions of this specification should be clearly set out in the offer.

Table (1)

Standard No.	Description	
IEC 60076 Power Transformers	Part 1	General
	Part 2	Temperature rise for liquid-immersed transformers
	Part 3	Insulation levels and dielectric tests
	Part 4	Lightning impulse and switching impulse testing
	Part 5	Ability to withstand short circuit
	Part 10	Determination of sound levels
	Part 11	Dry type power transformers
	Part 12	Loading guide for dry type power transformers
	Part 20	Energy Efficiency
IEC 60214 Tap-changers	Part 1	Performance requirements and test methods
	Part 2	Application guidelines
ISO 9001	International standard for a quality management system (QMS)	

3. ENVIRONMENTAL CONDITIONS

The electrical and mechanical properties of the required transformers should be guaranteed under the environmental conditions given in Table (2).

Table (2)

Ambient temperature	-5°C to +45°C (50°C as option according to …EDC) (35°C monthly average of the hottest month, 25°C yearly average) where the monthly and yearly averages are as defined in clause 3.12 of IEC 60076-1
Maximum relative humidity	95 %
Altitude	Up to 1000 m above sea-level

4. DESIGN CRITERIA

4.1 Rated Power

- The transformer should be able to deliver the maximum continuous power rating for an unlimited time at the main tapping of the tap changer without exceeding the allowable temperature rise limit, if the applied voltage is equal to the rated voltage and maintained at its rated frequency.
- The maximum continuous power rating is selectable from the set of standard rates given in Table (5).

4.2 Voltage Transformation Ratio

- The transformation ratio should be (22000/400 V) according to EDC requirement. The deviation of the transformation ratio measured at no-load and 50 Hz should not exceed 0.5% from its nominal value.
- The transformer should be able to work properly without damage as long as the voltage-to-frequency ratio does not exceed the corresponding ratio at rated voltage and rated frequency by more than 5%.

4.3 Vector Group

- The winding connection group of the transformers should be delta on the primary side and star on the secondary side in accordance with the vector group Dyn11 as specified in IEC 60076.

4.4 Tapping Range

- The taps are rigidly supported by cast into the epoxy along the central face of each high voltage coil; taps are changed by moving the flexible bolted links from one connecting point to the other, and tap terminals are clearly numbered and referenced on the transformer nameplate.
- Seven positions Tap-changer and the associated operating mechanism should be able to vary the transformation-ratio in steps of 2.5%. The range of transformation ratio should be +5% to -10%.
- The tap-changer enclosure should be equipped with a suitable transparent window of thickness not less than 3 mm to grant observation of the current tapping.
- Voltage tapping should be provided on the high voltage side of the transformer. Direction of operation for Raise/Lower tap should be provided.

4.5 On Load Tap Changer (Optional as per … EDC Requirements)

- In case of required on-load tap-changer (OLTC), the operating mechanism should be equipped with three-phase motor of 400 V and 50 Hz. All auxiliary supply voltage should be DC 110 V.
- The voltage sensing equipment should prevent operation of the tap-changer if the reference voltage is lost.
- The relays, switches and push buttons should be enclosed in a weatherproof cubicle and be readily accessible for maintenance at ground level with the transformer energized.
- The Motor Drive Unit of the OLTC should contain a thermostatically controlled anti-condensation heater, a door operated light and a manual handle for emergency and maintenance operation.

– It should be weatherproof and dust proof (IP54) and should be bolted to the transformer main tank in a convenient position such that the operator standing at ground level can carry out maintenance on all equipment contained in the cabinet with the transformer energized.

– Tap changer mechanism should send signal to initiate Indication/Alarm for the following:

1. Tap changer in progress
2. Tap changer incomplete
3. Out of step Indication
4. Supply voltage failure
5. Tap position
6. Tap changer counter

4.6 Overload Capacity

– Permissible overload capacity of air natural (AN) dry-type transformer should be in accordance to IEC 60076-12. For transformer with thermal class (F), the permissible overload capacity shall be as listed in Table (3).

Table (3)

Previous continuous loading (% of rated power)	Windings temperature/ hot spot	Duration of overloading for specific levels of overloading (% of rated power), Maximum temperature for hot spot 145 °C				
		10%	20%	30%	40%	50%
50 %	46 °C /54 °C	50 min	45 min	30 min	25 min	20 min
75 %	79 °C /95 °C	35 min	25 min	17 min	13 min	10 min
90 %	103 °C /124 °C	15 min	8 min	6 min	5 min	3 min
100 %	120 °C /145 °C	4 min	2 min	0 min	0 min	0 min

4.7 Short-circuit Withstand

– The transformer should be capable to withstand, without damage, the thermal and mechanical effects of short-circuit levels for the duration given in Table (4).

4.8 Temperature Rise Limits

– The transformer should be able to deliver its maximum continuous ratings without exceeding the following permissible temperature rise limits given in Table (4).

4.9 Insulation Level

– The test voltages given in Table (4) should be applicable to the transformer.

4.10 Impedance Voltage

– The short-circuit impedance (or impedance voltage Z% expressed as a percentage of rated voltage) measured at mid-tap at 95 ºC should be as follows:
  - 4 % for transformers of power rates (from 50 kVA up to 630 kVA)
  - 5 % for transformers of power rates (800 – 1000 – 1250 kVA)
  - 6 % for transformers of power rates (1500 – 1600 – 2000 – 2500 kVA)
  - 7 % for transformers of power rates (3150 – 5000 kVA)

– According to IEC 60076-1, the measured short-circuit impedance is subjected to a tolerance of ± 10% of the specified value.

Table (4)

Transformer kVA Rating		As given in Table (5) (1 p.u for AN / 1.4 p.u for AF)
Cooling Class		AN / AF
Type		Ventilated dry-type
Phases		3
Frequency		50 Hz
Taps		Off Load +2×2.5%, -4X2.5%, (OLTC is optional)(*)
Average winding temperature rise above ambient temperature at rated current "C"		95°C for 45°C ambient or 90°C for 50°C ambient (120°C is optional)(*)
Insulation system temperature		Class F, (Class H is optional)(*)
Ambient temperature		45°C, (50°C is optional)(*)
Primary Voltage		22 kV
Primary Connection		Delta
Secondary Voltage (at No load)		0.4 kV
Secondary connection		Star
Vector Group		Dyn11
Insulation level	Primary rated lightning impulse withstand voltage (kV_{peak})	125 kV
	Rated power frequency withstand voltage (kV_{rms})	50 / 5 kV
Duration of short circuit	Thermal (rms)	2 seconds
	Dynamic (peak)	0.5 s (Cat. 1), 0.25 s (Cat. 2) ±10%
S.C. level of 22 kV network		500 MVA
Impedance voltage at (75°C) at main tap		As given in Table (5)
Noise level		As given in Table (5)
Partial discharge		≤ 10 pC
Climate class		C1
Environmental class		E2
Fire protection		F1

(*) Option items as per ----EDC requirement.

4.11 Windings (Primary and Secondary Coils)

- The high and low voltage coils should be wound with high conductivity aluminum conductors (Copper is optional as per …. EDC requirements) and then solid-cast (encapsulated under vacuum and pressure) separately in molds.
- Coils should be separated by uniform air space that permits free circulation of cooling air between the coils.
- Windings should be properly sized, insulated and supported for the line and load voltages and currents to be encountered in operation including short circuits, when the transformer is properly installed using (class F), (class H is optional as per …. EDC requirements), rated insulated material and adequately protected.
- Construction of the transformer should separate the primary connections and the secondary connections by placing them on opposite sides of the core.
- The transformer core limb should be grounded to the base through a copper strap, and taps should be provided on the primary.

4.12 Encapsulation (for High- and Low-voltage Coils)

- The Encapsulation process should involve pre-heating the coils and then filling each mold with degassed and thoroughly mixed epoxy under vacuum. Once filled, the coils should then be oven cured to insure a solid void free casting. The oven curing process should be controlled and monitored to ensure that each coil is properly cured.
- The high voltage coils should withstand the half and full wave impulse tests as well as partial discharge tests to insure a properly cured and void–free casting.
- The epoxy should contain filler material providing characteristics superior to unfilled epoxy including higher temperature rating, better heat conductivity, better arc resistance and adhesion to the conductor, plus a coefficient of expansion closer to that of the conductor material.

4.13 Core

- The core should be constructed of the best quality, low loss; high permeability cold rolled grain-oriented steel laminations, on-aging silicon steel insulated on both sides.
- It should be miter-corner-cut without burs and hand stacked into a cruciform arrangement of different width of laminations, to provide a cross-section most nearly assembling the circular cross-section inside the low voltage coil.
- Laminations should be cut and stacked utilizing step-lap construction to reduce losses and sound.
- Designed flux density should be kept well below saturation **(≤ 1.7 Tesla).**
- The assembled core should be braced with heavy structural steel angle or channel to apply uniform clamping forces across the entire width, top and bottom.
- The core should have steel straps running top to bottom down the front and back of each core leg, inside the coils.
- Horizontal and vertical bracing should be through bolted to insure consistent clamping forces on all parts of the magnetic steel laminations.
- The core should be protected from corrosion with a higher temperature rated rust-resistant coating.
- The temperature of the core, metallic parts and adjacent materials should not reach a value that will cause damage to any part of the transformer.
- The core assembly should enable the removal of the coils in the field, if this should become necessary.
- The entire core assembly should be covered with a resin-based lacquer for corrosion protection before the coils are mounted.

4.14 Core and coil assembly

- Construction should consist of separate high and low voltage coil for each phase, mounted coaxially with high/low air space between coils adequate for the rated voltage potentials.
- Coils should be axially centered. It should be supported in its place on the transformer core

to restrain movement in either transportation or service.

- Each coil should be supported on epoxy or equivalent fiber glass blocks withstand mechanical and thermal strength, top and bottom, with resilient pads to retain the coils while permitting thermal expansion under loading.
- The core and coil assembly should be designed and manufactured to withstand, without damage, the short circuit testing as defined in IEC standards.

4.15 Busbar connection

- All bus work should be designed to withstand maximum short circuit stresses when connected to a supply system having a fault capacity of 20 kA for 3 seconds, symmetrical at rated voltage, and to meet all test voltage requirements, without audible or visible corona.
- Busbar should be sized to maintain a current density of less than or equal 1.5 A/mm^2 at full load. Suitable bimetal connectors AL/CU should be provided.
- Low voltage busbars connection should be high conductivity material, Connections of coil windings to busbar should be brazed or welded using shielded arc techniques or friction welding technique designed to provide high joint integrity. Connection and terminal bolting surfaces should be tin or silver-plated (for copper bars). Bolted connections should include spring washers to maintain bolted surface pressure. All busbars should be sized to conform to industry standards and practices, for the amperes involved and maximum specified ambient temperatures.
- High voltage busbar should be high conductivity material, and fixed with three epoxy resin HV insulators with 2 cm/kV min. Leakage path with bolt-connected joint and terminal surfaces tin or silver plated.
- High voltage leads delta connection should be copper busbar/rod/tube insulated with technically accepted types of terminations. The tap's leads should be flexible cable or solid links or jumpers.
- Ground busbar should be copper with 150 mm^2 C.S.A, extending the full length of two

opposite sides of the enclosure. It should have provisions to mount appropriate terminal connection for adjacent switchgear or external grounding.

– A galvanized shielded copper ground should be applied to facilitate the grounding of each metal part with the following specifications:
  - Copper: 99.99 % minimum purity.
  - Width and Thickness: 3 cm *1.5 mm at least. (effective C.S.A at least 35 mm^2)
  - Flexibility: full.
  - Number of Shielded conductors to ground the transformer body: 2

4.16 Enclosure (Optional as per … EDC Requirements)

– Enclosure's dimensions should be suitable for installation in transformer's room with IP23.
– Transformer with IP00 should be mounted in steel kiosk with IP23 to knockdown enclosure with anti-corrosion treatment (electrostatic painting).
– Enclosure should have a removable front face to gain access to the high voltage connection equipment with two easy handles.
– Entrance Cables: from up or/and down of the enclosure by reinforced aluminum material of thickness not less than 3 mm.
– Enclosure thickness for all sides, cover and bottom should be not less than 2.5 mm before electrostatic painting.
– Enclosures should be suitable for complete transformer lifting with Eye Hooks or Lifting Eyes.
– The factories should make the design and beams to approve that the total weight of the unit can be lifted from enclosure without removing the whole ceiling of it.
– The enclosure should be equipped with (3) suitable transparent windows of thickness not less than 3 mm (one for each phase), and should be against L.V. side of the transformer. (door is accepted with maintained IP23)
– The enclosure should be provided with two adequate fans to ensure uniform heat transfer (Option as per _ _EDC requirements).

4.17 Base

– The transformer base should be a separate rigid steel weldment capable of supporting the core-and-coil assembly with enclosure.
– Construction should include provisions for lifting as well as for jacking and skidding or rolling in both directions.
– Two stainless steel ground pads (nuts) with two tapped holes for attaching ground leads should be provided, one in each diagonally opposite corner of the base.
– Lifting provisions should be designed for lifting the transformer and the core-and-coil assembly, separately or together.

5. ACCESSORIES

– It is not necessary to approve the transformer's accessories separately. The approval issued for the transformer with the inclusions is sufficient.
– An instantaneous and maximum temperature – indicating thermometer should be provided on the front of transformer.
– The transformer and their accessories should be able to withstand the effect of the rated short circuit through windings for 2 seconds duration.
– The transformer and their accessories should be able to withstand the effect of the dynamic short circuit current through the windings for (0.5 s (Cat. 1) from 25 kVA up to 2500 kVA, 0.25 s (Cat. 2) from 2501 kVA up to 100 MVA) ±10%.
– The transformer should have a weatherproof nameplate of stainless-steel material placed in a clear position.

5.1 Thermal protection on LV windings and/or core

– Transformer should be provided with six fans to give a forced air-cooled rating above the self-cooled full rating, miniature circuit breaker and control wiring (wire markers included).
– A thermo-resistance PT100 (Platinum "100" for the resistance in ohms at 0 °C) should be connected to a suitable electronic device.

- PT100 thermo-resistance should allow logging of windings and core and a remote transmission of data.
- A thermal sensor and a fan controller should be supplied, installed inside the enclosure.
- The temperature controller should log the signal coming from three (3) PT100 sensors at least to indicate the measured temperature.
- It should include two relays with adjustable threshold for the alarm and release signals as well as an auxiliary contact to control a possible set of fans for air forced ventilation.
- It should be a microcontroller/microprocessor-based unit specifically designed for dry type and cast resin transformers.
- It should be developed with layout and advantages of the following technology platform:
    1. Dual display.
    2. Voting function.
    3. The controller with increased operational capacity.
    4. Data management.
    5. HFN (Hourly Fan No cycling)
- It should provide in a single product with the following outputs:
    1. Analog 4 – 20 mA
    2. Digital RS485 Modbus RTU.
- It should be able to monitor the temperature of the transformer by a remote communication system.
- It should in, addition to transmitting real-time temperatures, offer the possibility to vary all the settings including the relays intervention thresholds.
- It should be equipped with three (3) PT100 inputs (at least) to monitor the temperature of the core and windings while sensor inputs should be available.
- It should have the following:
    1. ALARM relays to give a signal for high temperatures.
    2. TRIP relay to give signal to disconnect the transformer in case it reaches the maximum threshold through C.B, RMU, ... etc.
    3. FANS relays to start the ventilation system.

4. FAULT relay to indicate the damage or break of PT100 sensors and the critical failures (trip operation failed).
5. SIGNAL relay for fans failure at BMS (Building Management System)

- It should be supplied with a coating on the electronic cards, resistant to difficult weather conditions, particularly characterized by high temperature and humidity.
- It should be universal power supply with input from 24 to 240 Vac/dc.
- "The fan controller box must be tightly closed to protect it from dust and impurities" such that the screen and the selector can be accessed from outside without need to open the enclosure.
- The thermal sensor and a fan controller should be used to indicate winding temperature and provide a single-phase voltage to operate fans or to disconnect fans according to the temperature of winding and auxiliary contacts to provide alarm and to trip the M.V.C.B.
- All control wiring and thermal sensor must be in ducts and thermal flexible plastic pipes.
- The manufacturer must provide a brief user manual (monitoring steps) for the fan-controller and to be installed beside it.
- The transformer should be provided with indicating lamps on the transformer outer surface with a lamp for each fan to indicate whether the fans are working properly or not, with an arrangement of the lamps indicating their dependency for each fan.

5.2 Anti-vibrating pads

- Anti-vibrating pads should be composed of special rubber supports supplied loose to be positioned by the … EDC under the transformer's wheels. (Noise Reduction Dumper)
- They should allow a great reduction of the vibrations transmitted to the structure and therefore of the noise and possibility of structural resonances.
- Anti-vibrating supports should be designed and supplied for special application in compliance with …EDC specification.

5.3 Ethernet communication channels (Optional)

- The Ethernet communication channel should be a microcontroller/microprocessor-based device.
- The Ethernet communication channel should be designed for dry type cast resin transformers with integrated Ethernet port.
- The Ethernet communication channel should be equipped with all the necessary functions needed to monitor and control the temperature of the transformer.
- The Ethernet communication with the network should be via Modbus TCP/IP protocol allowing to display and to program all the unit functions from the comfort of the desk.
- The Ethernet communication channel should maintain the traditional PT100 inputs and relays for ALARM, FAULT, TRIP, and FANs.
- The unit should be supplied with a coating on the electronic cards, resistant to difficult weather conditions, particularly characterized by high temperature and humidity.
- The protection relays should be universal power supply with input from 24 to 240 Vac / Vdc.

5.4 Ventilation system (set of electric fans for air forced (AF) cooling)

- Ventilation system should include two sets ionized aluminum bar cooling fans each bar consists of three fans, based on transformer type and power, fixed to the base of the transformer.
- To achieve its proper performance, each set of fans should be assembled together with a control and command switchboard.
- The cooling fans should provide optimal solutions for air forced cooling of dry transformers. The on-board ventilation system should maintain the transformer at its optimal temperature to increase the safety even in case of momentary overload. The structure of the fans should provide a fast and easy assembly / installation of the fans on the transformer enclosure.
- The cooling fans should have high performance crossflow and centrifugal to dissipate heat

generated by transformer losses while complying and assure fail-safe operations in all zones.

– The cooling fans should have the following specifications:

Operation	: indoor
Degree of protection	: IP ≥ 20
Speed	: ≥ 1800 rpm
Power	: ≤ 250 watt (for each fan)
Noise	: ≤ 71 dB
Installation	: vertical / horizontal
Ambient temperature	: -5 °C to 45 °C / +50 °C (optional)
Motor insulation class	: F

5.5 Junction box for PT100 sensors

– The cast resin transformer should be equipped with a box designed and manufactured for quick, safe and economic connection of PT100 sensors. The box should detect the temperature of the cast resin transformers.

– The box should be provided with three PT100 sensors at least with not less than 2.5 m cable for each sensor. The cable length as well as the sensor type should be determined according to the … EDC requirements.

– The box should have the following specifications:

Material	: Polyamide with cable gland features
Dimensions	: 10:14 mm
Degree of protection	: IP ≥ 54
Flame resistance	: UL 94V0 Operating
Operating temperature	: from -5°C to +45°C (+50°C is optional)

5.6 Data cables

– The data cables should be for the transport of the data signals with all the protection requirements: mod. CTES technical specifications of the extension cable for PT100.

– The data cables should provide the following specifications:

1 Cable 20 x AWG 20/19 Cu/Sn
2 Flame retardant insulation PVC105

3 CEI 20.35 IEC 332.1 regulations
4 Maximum operating temperature: 90°C
5 Conformation: 4 sets of three twisted and colored conductors
6 Shield in Cu/Sn
7 Flame retardant PVC sheath

5.7 Protection for temperature controller & fans protection

- This device should provide additional protection to protection units. The device should provide protection for the fans against over current and under voltage.
- This device should provide overvoltage protection and should be used on systems where there are large surges and fluctuations in 230 Vac power supply.
- It should be used for cutting the peak voltage generated when the MV switch is closed.
- It should be easily installed inside the panel through the DIN EN50022 rail.
- It should have a red led to indicate the possible intervention of fuses.
- It should have the following specifications:

Power supply	: 230 Vac.
Input voltage	: 230 Vac ± 10% 50Hz
Delay time	: settable from 1 to 5 seconds
Housing material	: Polyamide UL 94V2
Protection fuses	: 2 & 5 Amps (delayed)
Ambient operating temperature	: -5 °C to +45 °C (+50 °C is optional)

6. TRANSFORMER LOSSES

- The transformer should be designed for minimum losses.
- When comparing between different tenders, the present worth (PW) value of the annual capitalized cost of guaranteed losses in the transformers should be added according to the following formula:

$$PW = K \times 8760 \times C\ [W_i + (LSF)\ (P^2)\ W_{cu}]$$

Where

K = Present worth factor (L.E)

$$K = ((1+i)^N - 1) \,/\, i\,(i+1)^N\,)$$

i: latest interest rate from Central Bank of Egypt.

N: number of years. (20 years)

C = Cost of kWH. (According to the max commercial cost of kWH and can be adjusted as per EEHC latest update)

LSF = Loss factor for the load = 0.4.

W_i = Iron (no-load) losses in kW at rated voltage and main tapping

W_{cu} = Copper (load) losses in kW at full load and 95 ºC.

P = Peak load (P.U.) = 0.8.

– The values of the required losses are given in Table (5)

Table (5)

Power (kVA)	No load losses (W)	Load losses (W) at 75ºC	% Z at 75ºC (±10%)	Noise Level dB
50	380	1200	4	45
63	460	1400	4	48
100	590	1700	4	50
160	650	2300	4	50
250	880	3300	4	53
315	1030	4000	4	54
400	1200	4800	4	55
500	1500	5600	4	56
630	1650	6800	4	57
800	2000	8200	5	58
1000	2200	8900	5	59
1250	2800	11500	5	61
1500	2800	12800	6	61
1600	3100	14000	6	61
2000	4000	17500	6	62
2500	5000	20000	6	65
3150	6300	23000	7	65
5000	10000	31500	7	65

7. MARKING

- Transformer serial No. and rated power should be engraved on transformer body at LV side.
- Each transformer should be fitted with a nameplate of weather proof materials fitted in a visible position, showing the information listed below. Entries on the plate should be indelibly marked by etching or engraving:

1. Type of transformer	2. IEC codes	3. Manufacturer's name
4. Serial number	5. Year of manufacture	6. Insulation level
7. Number of phases	8. Rated power	9. Rated frequency
10. Rated voltage	11. Rated current	12. Vector group
13. Tap position	14. Impedance voltage (% Z)	15. Type of cooling
16. Total mass	17. Temperature rise of windings	18. Customer: … EDC
19. Short Circuit Duration	20. P.O No. / Year	

- If the enclosure is requested, a second similar nameplate should be fixed by rivets on a visible position of the outside surface of the enclosure.

8. TESTING and INSPECTION

8.1 General

- Tests shall be carried out on finished transformers in accordance with the latest relevant standard and as specified herein.
- All tests shall be carried out by and at the expense of the tenderer who shall supply all the apparatus, instruments and equipment.
- Routine tests shall be carried out in the presence and under the control of the inspecting engineer of _ _ EDC, except in specified or approved cases where test certificates are accepted. All test results shall be reviewed and approved in writing by _ _ EDC.

- The contractor should advise testing dates. The contractor shall not be entitled to any extension of the time of completion due to the failure of any test or the rejection of any part of the materials as a result of any test.
- Routine tests shall be carried out in the supplier's factory or at any other mutually approved place.

8.2 Type Tests:

- Type tests should be carried out on one transformer for each type. Type tests should include and not limited to the following:

1-	Full wave lightning impulse voltage withstand test.	IEC 60076
2-	Temperature rise test.	IEC 60076

8.3 Special Tests

he special tests shall be according to IEC standards for dry type transformers. These are including and not limited to the following:

1. Measurement of sound level test
2. Environment test.
3. Climatic test.
4. Fire behavior test.
5. Short Circuit test

- The supplier shall submit short circuit test in a approved lap by the EEHC , otherwise supplier shall comply with the following:
  a. Short circuit calculations for transformer need approval shall be submit to Extra High Voltage Research Center for review with same sample transformer with its drawing.
  b. Supplier shall be guarantee transformer for 3 years, guarantee will continues to be vailed for this period or till submit short circuit test whichever is earlier
  c. Supplier will get approval for one year only and renewed yearly if there is no complain form any --EDC during guarantee period (3 years)

8.3 Routine Tests:

- Routine tests should be applied on each transformer in accordance with the requirements of IEC 60076-11 standards for dry type transformer. Routine tests should include and not limited to the following:
  1- Measurement of winding resistance.
  2- Measurement of voltage ratio and check of phase displacement
  3- Measurement of short-circuit impedance and load loss
  4- Measurement of no-load loss and current.
  5- Applied voltage dielectric test.
  6- Induced A.C withstand voltage test.
  7- Partial discharge measurement (the value should be less than 10 pC

8.5 Penalty and Rejection

8.5.1 Rejection of material:

- Failure of any material to comply with any one of the requirements of this specification shall constitute grounds for rejecting that material, except if the ___EDC inspection engineer convinced that the material could satisfactorily be repaired. In such case, the contractor should be obliged to replace the defective material without being entitled to any extra payment or to any extension of time to complete the contract.

8.5.2 Penalty and rejection for decreased efficiency:

- The contractor should keep the no-load losses and load-losses to be within the guaranteed values.
- Should the measured losses exceed the allowable tolerances, then ___EDC representative should reject the transformer with such losses. However, if the losses deviation lies within the allowable tolerances (10 % of the total losses), then the excess losses should be evaluated as per the following formula and the contractor should be penalized accordingly:

$$\mathbf{PW = K \times 8760 \times C\,[dW_i + (LSF)\,(P^2)\,dW_{cu}]} \qquad \textbf{LE}$$

Where

K = Present worth factor (L.E)

$K = ((1+i)^N - 1) / i\ (i+1)^N)$

i: latest interest rate from Central Bank of Egypt.

N: number of years. (20 years)

C = Cost of kWH. (According to the max commercial cost of kWH and can be adjusted as per EEHC latest update)

LSF = Loss factor for the load = 0.4.

dW_i = Excess in iron (no-load) losses in kW at rated voltage and main tapping

W_{cu} = Excess in copper (load) losses in kW at full load and 95 °C.

P = Peak load (P.U.) = 0.8.

– No credit should be given for any decrease in the losses below the guaranteed values.

8.5.3 Replacement of rejected material and equipment

– In case of any rejected material or equipment, the contractor should be obliged to replace it without extra payment.

9. SUBMITTALS

9.1 Submittals required with tender:

– The tenderer shall complete and return one legible copy of the attached Data Schedule for each rate of the offered transformers.

– Dimensional drawings of the offered transformers including interconnection diagrams of all components and assemblies.

– Original catalogues shall be submitted to facilitate evaluation of the offer.

– All information should be in English language, clearly readable, with IEC standard symbols and SI units.

– Type and special test reports/certificates from an independent testing laboratory shall be submitted to _ _ EDC.

– Approval certificate issued by Egyptian Electricity Holding Company.

- List of deviations & clauses to which exception is taken (if any).
- **Spare parts list** (option as per ...EDC requirements)

  The tenderer should submit a list of recommended spare parts.

9.2 Submittals required following award of contract:

- The contractor should provide with shipment of the transformer complete installation, operation and maintenance instruction/manuals suitable for the specific supplied item(s), including information from the manufacturer of each purchased accessory.
- It should include interconnected diagrams, drawings and instructions describing procedures for items that will be shipped to be assembled and interconnected for operation when installed.
- These should contain strict instructions as required for carrying, moving by skidding or rolling, and lifting the transformer and accessories.
- The manual should include the instructions for tap changing, enclosure assembly, disassembly, temperature indicator reading, setting procedures and other information designed to assist in proper operation and periodic maintenance of the supplied equipment.
- Key interlocking should be identified, and serial number should be supplied.
- All manuals, instructions and supporting drawings and parts lists supplied by the manufacturer and supplemental instruction and data supplied by accessory manufacturers should be in English, clearly readable and arranged in such format to make the subject items easily identifiable.
- The contractor should provide a complete set of “as manufactured” drawings and wiring diagrams including interconnection diagrams of all components and assemblies of the equipment which assist in operation and maintenance of the equipment.
- All drawings should be compatible with current industrial standards and good industrial practice.
- All information should be in English language, clearly readable, with IEC standard symbols, device model, numbers and other requirements.

- All Dimensions should be in centimeters.

10. GUARANTEE

- The supplier guarantee the transformer against all defects arising out of faulty design or workmanship, or of defective material for a period of 12 months from putting in service or 18 months from delivery date.
- In case of supplying for the first time, the guarantee period should be extended to 4 years at least from delivery date.

11. TECHNICAL DATA SCHEDULE

- The tenderer must fill in thoroughly the attached technical data schedule.
- Any offer does not accompanied with clear and complete technical data schedule shall be rejected.

TECHNICAL DATA SCHEDULE
Dry TRANSFORMER 22 kV / 0.4 kV

- Manufacturer	....
- Type of transformer	....
- Method of cooling	....
- Standard of specifications	....
- Vector group - Winding Techniques: Primary Secondary	....
- Transformer ratio at main tapping and no load	.... / ....kV
- Transformer ratio at main tapping and full load:	.... / ....kV
- At unity P.F.	.... / ....kV
- At 0.8 P.F.	.... / ....kV
- Rated primary current at main tapping and full load	..................A
- Rated secondary current at main tapping and full load	..................A
- Current density in primary winding at full load	. ...A/mm^2
- Current density in secondary winding at full load	. ...A/mm^2
- Primary current at main tapping and no load	..................A
- Flux density in core and yoke at rated voltage at main tapping and full load	................Tesla
- Rated output with self-air cooling	....................kVA
- Rated output with forced air cooling	....................kVA
- Permissible overload duration without changes of specified temperature rise	
- 10% overload starting from 50% rated output	..................hr
- 20% overload starting from 75% rated output	..................hr
- 30% overload starting from 90% rated output	...................hr
- 40% overload starting from 100% rated output	..................hr
- 50% overload starting from 100% rated output	..................hr
- Tap changer (On load / Off load)	.... / ....
- Number of steps	....
- Voltage corresponding to each step.	..................kV
- Insulation levels for line terminal	
- Impulse withstand voltage + 1/50 microseconds wave (peak)	..................kV
- One-minute power frequency withstand voltage for primary and secondary windings (RMS)	..................kV

- Max symmetrical short circuit stresses withstand at rated voltage	……………….kA
- Induced voltage test for primary winding (RMS)	……………….kV
- Electrical performance and characteristic	
- Resistance per phase at 75 °C - Primary winding	……………….Ω
- Secondary winding	……………….Ω
- Impedance voltage at full load and main tapping 75 °C	……………….Ω
- Equivalent resistance referred to primary winding	……………….Ω
- Equivalent reactance referred to primary winding	……………….Ω
- Regulation at	
- Unit PF and 4/4, 3/4, 1/2, 1/4 Full load	………………….%
- 0.8 PF lagging 4/4, 3/4, 1/2, 1/4 Full load	………………….%
- magnetizing primary current at main tapping	………………….A
- Partial discharge	………………….pC
- Losses and efficiency	
- Load losses at full load and main tapping at unity PF and 75°C	……………….kW
- Iron losses at rated voltage and main tapping	……………….kW
- Total losses at full load and main tapping at unity PF and 75°C	……………….kW
- Efficiency at	
140% full load at 0.8 / unity PF	………………….%
125% full load at 0.8 / unity PF	………………….%
100% full load at 0.8 / unity PF	………………….%
75% full load at 0.8 / unity PF	………………….%
50% full load at 0.8 / unity PF	………………….%
25% full load at 0.8 / unity PF	………………….%
- Max temperature rise at 45°C ambient at full load of	
Winding	…………………. °C
Iron core	…………………. °C
Hot spot	…………………. °C
- Insulation class	…………………..
- Cooling system	…………………….
- Number of cooling units	………………….
- Number of air fans per cooling units fan diameter /rating / and discharge No./mm/HP/1tr/min	…………………….
- Material of core	………………….

- Material of Enclosure	……………………
- Thickness of Enclosure - Sides - Cover - Bottom	 ………………..…mm ………………..…mm ………………..…mm
- Insulation material - Primary winding - Secondary winding - Major insulation - Core plates	 ………………. ………………. ………………. ……………….
- Number and type of coils per phase: - Primary windings (AL/CU) - Secondary windings (AL/CU)	………………. ………………. ……………….
- Weight of core	……………….kg
- Weight of windings	……………….kg
- Weight of transformer ready for shipping	……………….kg
- Dimension of transformer (without enclosure) (L× W× H)	…………………..…
- Dimension of enclosure (L× W× H)	…………………..…
- The minimum clearance from winding to enclosure	…………………..…
- Protection degree (IP) for: Cooling fans Enclosure Junction box	 ………………. ………………. ……………….
- Noise level	……………….
- Climate class	……………….
- Environmental class	……………….
- Fire protection class	……………….

I/we guarantee the correctness of above data for transformers we are offering to Electricity Distribution Company.

Signature of tenderer: ……………..   Date: /   /